Mohith Ram Narendra Babu

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Education

University of California, Berkeley — Berkeley, CA

B.S. in Electrical Engineering & Computer Sciences — *Expected Graduation: May 2028*

- **Relevant Coursework:** Structure & Interpretation of Computer Programs, Data Structures, Efficient Algorithms, Linear Algebra, Circuits, Signals & Systems, Discrete Math, Probability, Machine Structures, Machine Learning

Professional Experience

Stellantis — Berkeley, CA

Software Engineer [Contract] — *September 2026 – Present*

- Build reservation-gated SSH via OpenSSH AuthorizedKeysCommand, admitting user keys only during active bookings
- Develop systemd heartbeat agent posting load and active users every 30s over outbound-only HTTPS to a status API
- Design PostgreSQL/Prisma reservation engine with transactional conflict checks and waitlists to prevent double-booking
- Build containerized mock bench fleet and OIDC provider to test SSH, SSO, and 3-tier RBAC end to end across 7 labs

Fortinet — Berkeley, CA

Technical Consultant [Contract] — *August 2026 – Present*

- Build ranking pipeline scoring 100K+ accounts on funding, hiring, and sector-growth signals, cutting latency 70%
- Develop growth-potential model across 20+ behavioral and market features, raising high-growth precision 25%
- Cut end-to-end batch inference time 70% across 10K+ accounts by parallelizing feature computation and scoring
- Build monitoring platform serving model outputs via FastAPI, PostgreSQL, and React analytics, cutting reporting 60%

Amazon Web Services (AWS) — Seattle, WA

Software Development Engineering Intern — *June 2026 – August 2026*

- Cut agreement lookup time to <2s by replacing manual search with a Python Bedrock agent mapping queries to 5 APIs
- Caught 80% of regressions pre-deployment by replaying 61 labeled queries across 10 workflows with benchmark harness
- Eliminated write risk with a read-only API allowlist, rejecting non-GET calls pre-dispatch and enforcing with 22 tests
- Reduced lookup tickets 45% across 5 teams by shipping retrieval and API-orchestration workflows on Linux hosts

Maserati — Berkeley, CA

Data Consultant [Contract] — *February 2026 – July 2026*

- Built benchmarking model across 15+ luxury brands, identifying EV and pricing gaps to guide 5–10 year roadmap
- Analyzed 10K+ consumer reviews with NLP models, quantifying perception gaps in innovation vs. heritage positioning
- Automated dashboards tracking electrification, AI, and sustainability trends, cutting research time 40%

Projects

Vandal: Self-Healing Guard for AI Finance Agents | *Python, FastAPI, Next.js, Claude API* — October 2026

- Built finance-agent guard with CEGIS-synthesized controls, stopping 40/40 held-out fraud attacks with 0 false blocks
- Wrote bounded BFS attack search with state-hash dedup + partial-order reduction, re-verifying 34K+ sequences
- Designed deterministic event-sourced financial twin (integer cents, sim clock), enabling exact replay and delta-debugging

Kalshi Mispricing Engine | *Rust, Tokio, WebSockets* — August 2026

- Ingested 12K+ NFL order-book updates/sec via WebSocket using concurrent Tokio pipeline w/ lock-free async channels
- Surfaced 80+ mispricings/week by comparing live prices to win probabilities and flagging $>2\sigma$ deviations
- Backtested strategy at 14.7% simulated return across 1,200 historical games, modeling transaction fees and slippage

GPT-2 Inference Engine | *C++17, Multithreading* — July 2026

- Implemented GPT-2 inference in C++17, achieving 42 tok/s on CPU with zero ML-framework dependencies
- Cut per-token latency 61% using KV caching, cache-blocked GEMM kernels, and multithreaded attention across 8 cores
- Verified logits within 1e-4 of HuggingFace outputs across 100 prompts using an automated differential-testing harness

Skills

Languages: C++, C, Rust, Python, Java, TypeScript, JavaScript, SQL, RISC-V Assembly, Bash
Systems & ML: Multithreading, SIMD, OpenMP, Async I/O (Tokio), KV Caching, GEMM Kernels, NLP, LLM Agents
Backend & Web: FastAPI, Pydantic, Node.js, Next.js, React, PostgreSQL, Prisma, REST APIs, OIDC/SSO, RBAC
Infra & Tools: Linux, Docker, systemd, OpenSSH, Git, GitHub Actions, pytest, AWS (Bedrock, CDK, CloudWatch)